Outpatient Imaging Efficiency: Abdomen Computed Tomography (CT) Use of Contrast Material Measure Performance

Qualified Entity Public Report

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OVERVIEW

Acumen, LLC conducts policy research in support of federal, state, and local healthcare and social policy programs. Our team leverages its vast data resources to improve information available to policymakers on topics such as quality measurement, payment policy, drug and vaccine safety, health insurance markets, value-based purchasing (VBP), and healthcare fraud, waste, and abuse (FWA). Our work reflects the belief that policymakers, healthcare providers, and the public should have the best available information upon which to base their decisions.

Acumen produced this report as part of its participation in the Centers of Medicare & Medicaid Services (CMS) Qualified Entity (QE) program. Under the QE Certification Program (QECP), CMS provides QEs with standardized extracts of Medicare Parts A and B claims data and Part D prescription drug event (PDE) data to be combined with those from other payer data sources to generate reports that give insights into quality performance. For more information, visit https://www.qemedicaredata.org/.

The purpose of this report is to provide insight on the use of abdomen or abdomen and pelvis (abdominopelvic) computed tomography (CT) scans with and without contrast in hospital outpatient settings. The appropriate use of imaging is important for improving patient safety and healthcare efficiency. Professional organizations have launched initiatives to reduce the unnecessary use of imaging procedures. For example, the American College of Radiology (ACR) publishes evidence-based guidelines for appropriate use of imaging in a variety of clinical scenarios to assist clinician decision-making and encourage more efficient use of imaging.[1] The American Board of Internal Medicine (ABIM) Foundation's Choosing Wisely initiative encourages conversations among clinicians, providers, and patients about whether tests or other services are medically necessary.[2] The Medicare Payment Advisory Commission (MedPAC) identifies a substantial share of low-value care (i.e., services that have little or no clinical benefit or care and in which the risk of harm from the service outweighs its potential benefit) as imaging services.[3]

This report contains results from the Abdomen Computed Tomography (CT) Use of Contrast Material measure. It was calculated with QE-Medicare Fee-for-Service (FFS) claims data from CMS and commercial claims data. Section 1 of this report describes the Abdomen CT Use of Contrast Material measure and its specifications, and Section 2 displays the measure results.

[1] American College of Radiology, ACR Appropriateness Criteria. https://www.acr.org/Clinical-Resources/Clinical-Tools-and-Reference/Appropriateness-Criteria (last accessed February 10, 2025)
[2] ABIM Foundation, Choosing Wisely. https://www.choosingwisely.org/ (last accessed February 10, 2025)
[3] Medicare Payment Advisory Commission. 2024. "A Data Book: Health Care Spending and the Medicare Program." https://www.medpac.gov/wp-content/uploads/2024/07/July2024_MedPAC_DataBook_SEC.pdf.

1 MEASURE DESCRIPTION

The Abdomen CT Use of Contrast Material measure calculates the percentage of abdomen CT scans that are performed without and with contrast (i.e., a combined abdomen CT scan) out of all of the abdomen CT scans performed in the outpatient setting—those without contrast, those with contrast, and those without then with contrast. Abdomen CTs include CT scans of the abdomen and of the abdomen and pelvis.

The following sections describe the Abdomen CT Use of Contrast Material measure.

1.1 Measure Use in CMS Programs

This measure is stewarded by CMS and has been used in the CMS Hospital Outpatient Quality Reporting (HOQR) program since 2010 and in the Rural Emergency Hospital Quality Reporting (REHQR) program since 2024. The HOQR and REHQR programs collect and report quality measure data from short-term acute care hospitals and REHs, respectively, for hospital outpatient services paid under the Outpatient Prospective Payment System (OPPS). Detailed measure specifications are available on the Hospital Outpatient Imaging Efficiency Measures CMS web page.[4]

This measure aims to promote use of high-quality, efficient care; reduce unnecessary exposure to radiation and contrast materials; and ensure adherence to evidence-based medicine and clinical practice guidelines.

1.2 Denominator

The initial population for the measure is all patients who underwent an abdomen CT scan (without contrast, with contrast, or combined) at an outpatient facility. Each abdomen CT scan at an outpatient facility is counted once in the measure's denominator. Patients who had more than one abdomen CT study in the measurement period can, therefore, be included in the measure's initial patient population multiple times. This measure does not include patients who were admitted to the hospital as inpatients.

1.3 Numerator

The numerator is the number of patients with a combined abdomen CT scan.

1.4 Excluded Conditions

Patients who have a clinical diagnosis of one or more conditions for which combined abdomen CTs are considered appropriate are excluded from the measure. As a result, this measure excludes patients with any the following conditions:

[4] CMS, Hospital Outpatient Imaging Efficiency Measures. https://qualitynet.cms.gov/outpatient/measures/imaging-efficiency (last accessed February 10, 2025)

- Adrenal mass
- Hematuria
- Infections of the kidney
- Jaundice
- Liver lesion (mass or neoplasm)
- Malignant neoplasm of the bladder
- Malignant neoplasm of the pancreas
- Diseases of the urinary system
- Pancreatic disorders
- Non-traumatic aortic disease
- Unspecified disorders of the kidney or ureter

1.5 Measurement Period

The measure is calculated using a one-year period.

1.6 Measure Calculation

This measure calculates the percentage of abdomen CT scans that are performed with and without contrast out of all abdomen CT scans (those without contrast, those with contrast, and those with both combined) in the hospital outpatient setting.

The interpretation of measure results is that lower rates indicate better performance. That is, a facility with better performance will have a rate closer to 0%, whereas a facility that may be performing too many combined CTs of the abdomen will have a higher rate.

1.7 Data and Study Period

The measure is calculated using Medicare FFS claims data from CMS and commercial claims data for calendar years (CY) 2020 and 2021 for all 50 states and the District of Columbia (DC).

2 MEASURE RESULTS

This section summarizes the results for the Abdomen CT Use of Contrast Material measure for CY 2020 and 2021. Section 2.1 shows the percentage of combined abdomen CT scans at the national and state levels, Section 2.2 examines combined abdomen CT rates by patient age and sex, and Section 2.3 provides details on the frequency of the excluded conditions. Finally, Section 2.4 displays abdomen CT counts and combined abdomen CT rates by month.

2.1 Measure Results at the National and State Levels

Table 1 displays the measure results at the national level for CY 2020 and 2021.

Table 1. National Combined Abdomen CT Use of Contrast Material Rates

	Medicare FFS + Commercial	Medicare FFS	Commercial
CY 2020			
Denominator	5,594,709	2,195,646	3,399,063
Numerator	277,187	138,339	138,848
Rate	4.95%	6.30%	4.08%
CY 2021			
Denominator	5,770,512	2,368,594	3,401,918
Numerator	276,629	145,851	130,778
Rate	4.79%	6.16%	3.84%

Figures 1 and 2 show the measure results at the state level. In CY 2020, Louisiana and Alabama had the highest rates of combined abdomen CTs at 9.98% and 8.71%, respectively. Nevada and Minnesota had the lowest rates at 2.53% and 2.95%, respectively. In CY 2021, Louisiana and Alabama also had the highest rates of combined abdomen CTs at 10.06% and 8.82%, respectively. Maryland and Nevada had the lowest rates that year at 1.60% and 2.06%, respectively.

Figure 1. Measure Results by State, CY 2020

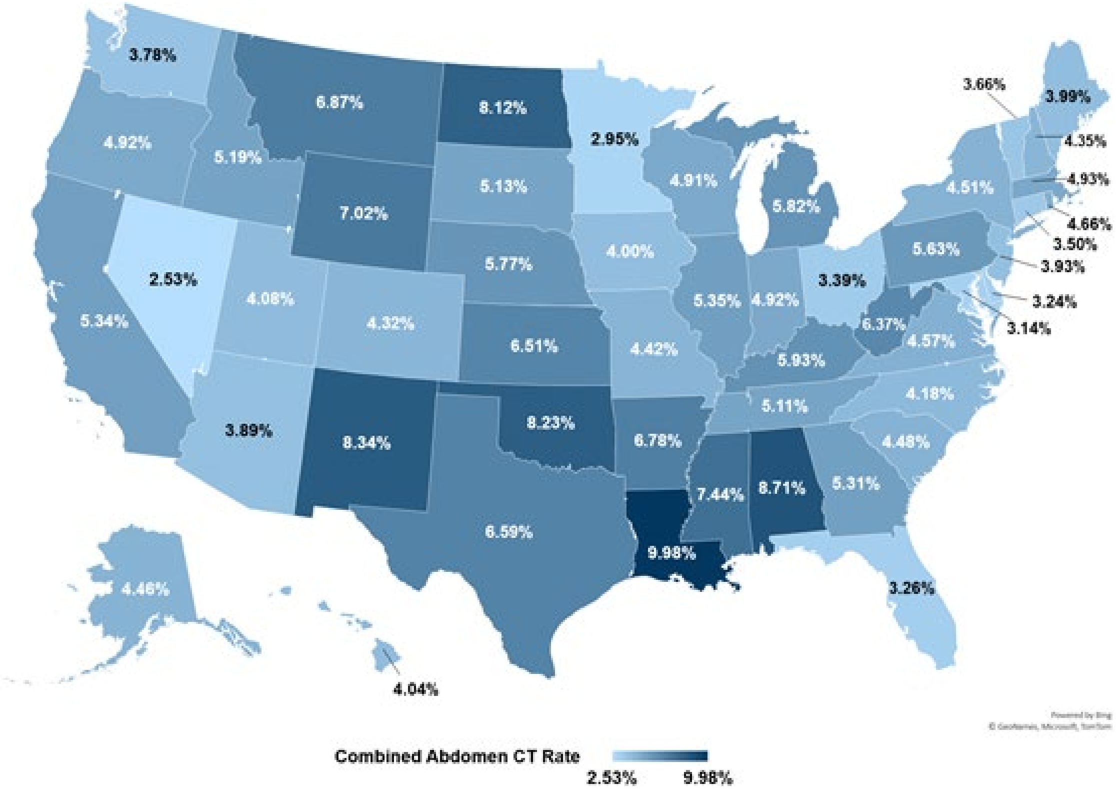
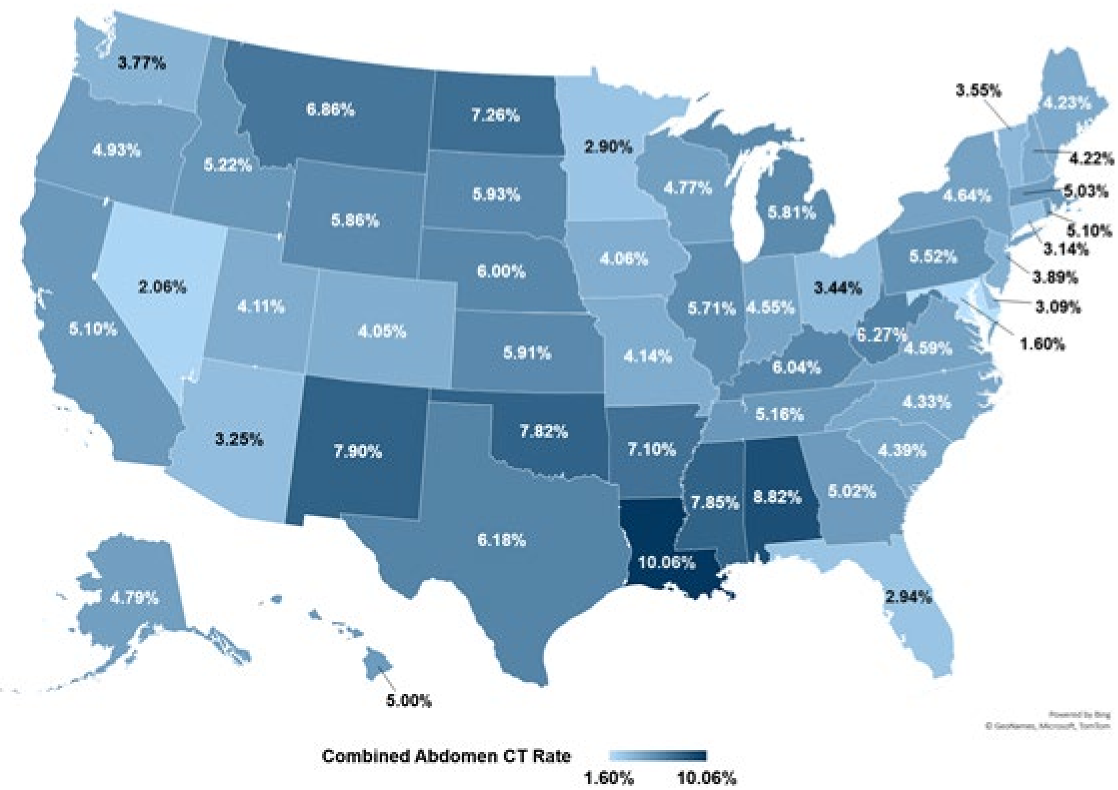


Figure 2. Measure Results by State, CY 2021



2.2 Measure Results by Patient Characteristics

Acumen calculated combined abdomen CT rates by patient characteristics.[5] Table 2 displays rates for patients age under 65 and those age 65 and older. Rates among patients age 65 and older were higher than the rates among patients age under 65 in both years. Table 3 shows rates for female patients and male patients. The combined abdomen CT rates among male patients were higher than among female patients in both years. All differences in rates by age and by sex were statistically significant at the one percent level.[6]

Table 2. Measure Results by Patient Age

	Medicare FFS + Commercial			Medicare FFS			Commercial		
	Total	**Under 65**	**65 and Older**	**Total**	**Under 65**	**65 and Older**	**Total**	**Under 65**	**65 and Older**
CY 2020									
Denominator	5,542,135	2,774,984	2,767,151	2,195,646	430,292	1,765,354	3,346,489	2,344,692	1,001,797
Numerator	275,709	92,468	183,241	138,339	18,441	119,898	137,370	74,027	63,343
Rate	4.97%	3.33%	6.62%	6.30%	4.29%	6.79%	4.10%	3.16%	6.32%
CY 2021									
Denominator	5,716,283	2,727,132	2,989,151	2,368,594	410,375	1,958,219	3,347,689	2,316,757	1,030,932
Numerator	275,142	83,983	191,159	145,851	17,287	128,564	129,291	66,696	62,595
Rate	4.81%	3.08%	6.40%	6.16%	4.21%	6.57%	3.86%	2.88%	6.07%

Table 3. Measure Results by Patient Sex

	Medicare FFS + Commercial			Medicare FFS			Commercial		
	Total	**Female**	**Male**	**Total**	**Female**	**Male**	**Total**	**Female**	**Male**
CY 2020									
Denominator	5,593,375	3,191,947	2,401,428	2,195,646	1,240,444	955,202	3,397,729	1,951,503	1,446,226
Numerator	277,123	137,667	139,456	138,339	67,457	70,882	138,784	70,210	68,574
Rate	4.95%	4.31%	5.81%	6.30%	5.44%	7.42%	4.08%	3.60%	4.74%
CY 2021									
Denominator	5,769,359	3,331,190	2,438,169	2,368,594	1,352,528	1,016,066	3,400,765	1,978,662	1,422,103
Numerator	276,595	138,417	138,178	145,851	71,806	74,045	130,744	66,611	64,133
Rate	4.79%	4.16%	5.67%	6.16%	5.31%	7.29%	3.84%	3.37%	4.51%

2.3 Frequency of Excluded Conditions

Acumen analyzed the frequency of excluded conditions for CY 2020 and 2021. Table 4 displays the counts of abdomen CT scans for each excluded condition, the total excluded CT scans, and the non-excluded CT scans (i.e., the measure denominator). The table also displays the percentages of each group relative to the initial population. The most common excluded

[5] Cases with missing age or sex information were excluded from the counts in Tables 2 and 3.
[6] Statistical significance is based on the chi-square proportion tests.

conditions across data sources were diseases of the urinary system and hematuria, accounting for approximately 55% of excluded cases.

Table 4. Frequency of Excluded Conditions

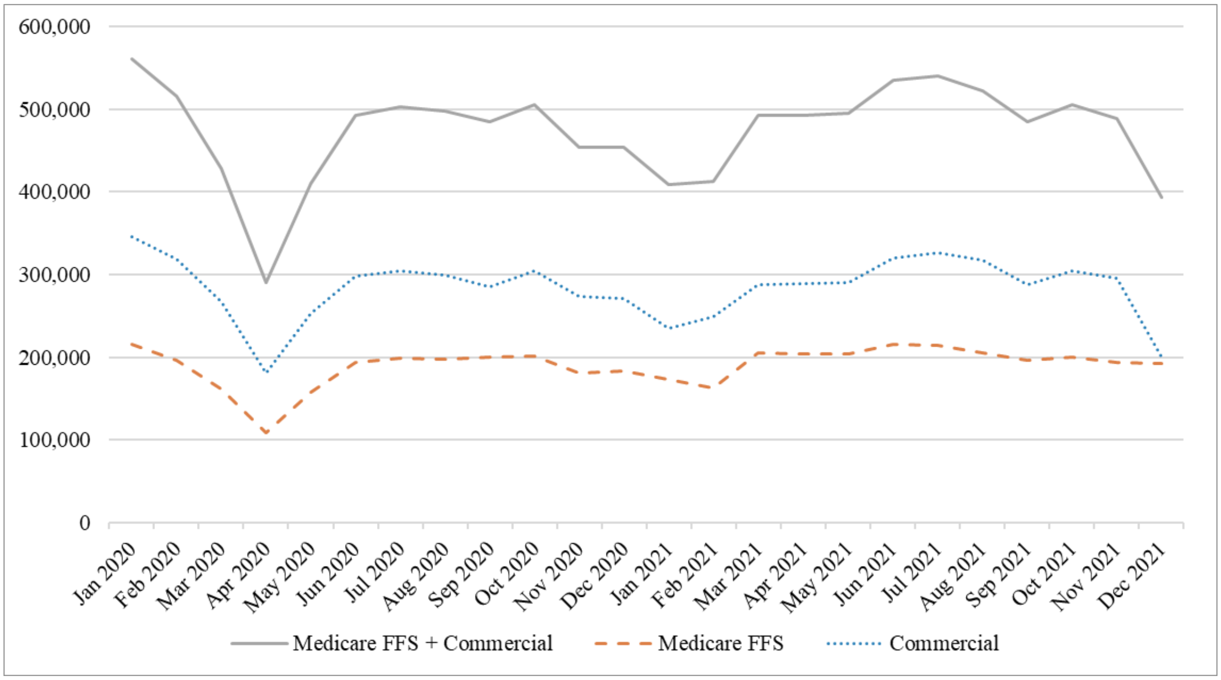
Diagnostic Categories	Medicare FFS + Commercial		Medicare FFS		Commercial	
	N Cases	% Cases	N Cases	% Cases	N Cases	% Cases
CY 2020						
Initial Population (Cases)	**7,050,186**	**100.00%**	**2,880,692**	**100.00%**	**4,169,494**	**100.00%**
Adrenal mass	30,080	0.43%	16,056	0.56%	14,024	0.34%
Diseases of the urinary system	460,695	6.53%	183,841	6.38%	276,854	6.64%
Hematuria	328,995	4.67%	152,120	5.28%	176,875	4.24%
Infections of kidney	74,632	1.06%	17,204	0.60%	57,428	1.38%
Jaundice	9,163	0.13%	4,391	0.15%	4,772	0.11%
Liver lesion (mass or neoplasm)	114,126	1.62%	63,940	2.22%	50,186	1.20%
Malignant neoplasm of bladder	129,964	1.84%	77,826	2.70%	52,138	1.25%
Malignant neoplasm of pancreas	83,056	1.18%	47,788	1.66%	35,268	0.85%
Non-traumatic aortic disease	197,293	2.80%	122,216	4.24%	75,077	1.80%
Pancreatic diseases	91,265	1.29%	34,630	1.20%	56,635	1.36%
Unspecified disorder of kidney and ureter	64,208	0.91%	34,194	1.19%	30,014	0.72%
Total Excluded Population*	**1,455,477**	**20.64%**	**685,046**	**23.78%**	**770,431**	**18.48%**
Measure Denominator	**5,594,709**	**79.36%**	**2,195,646**	**76.22%**	**3,399,063**	**81.52%**
CY 2021						
Initial Population (Cases)	**7,254,273**	**100.00%**	**3,094,271**	**100.00%**	**4,160,002**	**100.00%**
Adrenal mass	28,458	0.39%	15,409	0.50%	13,049	0.31%
Diseases of the urinary system	496,513	6.84%	207,152	6.69%	289,361	6.96%
Hematuria	333,562	4.60%	164,301	5.31%	169,261	4.07%
Infections of kidney	77,376	1.07%	19,093	0.62%	58,283	1.40%
Jaundice	9,598	0.13%	4,645	0.15%	4,953	0.12%
Liver lesion (mass or neoplasm)	106,653	1.47%	62,476	2.02%	44,177	1.06%
Malignant neoplasm of bladder	130,867	1.80%	81,683	2.64%	49,184	1.18%
Malignant neoplasm of pancreas	81,492	1.12%	48,611	1.57%	32,881	0.79%
Non-traumatic aortic disease	197,344	2.72%	125,634	4.06%	71,710	1.72%
Pancreatic diseases	89,729	1.24%	34,912	1.13%	54,817	1.32%
Unspecified disorder of kidney and ureter	61,769	0.85%	33,922	1.10%	27,847	0.67%
Total Excluded Population*	**1,483,761**	**20.45%**	**725,677**	**23.45%**	**758,084**	**18.22%**
Measure Denominator	**5,770,512**	**79.55%**	**2,368,594**	**76.55%**	**3,401,918**	**81.78%**

* Exclusions are not mutually exclusive. Therefore, the sum of individual excluded conditions is greater than the total number of excluded cases.

2.4 Measure Results by Month

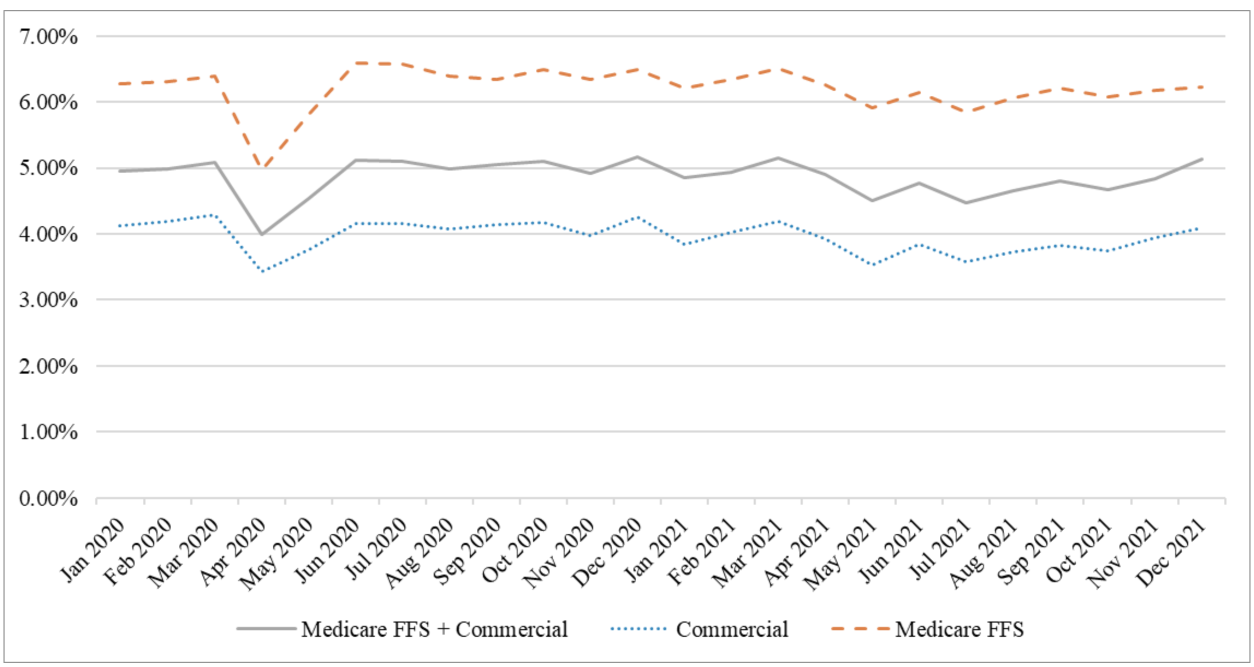
Figures 3 and 4 provide the counts and rates of combined abdomen CTs, respectively, over time. The number of abdomen CT scans per month remained fairly stable over time, with the most notable change in counts occurring in early 2020. The largest month-to-month decrease was from March 2020 to April 2020 when the count dropped from 428,502 to 289,659 scans. This was a short-term decrease: in June 2020, the number of scans increased to 492,291.

Figure 3. Monthly Abdomen CT (Denominator) Counts [7]



Similarly, the rate of combined abdomen CT scans over total abdomen CT scans remained fairly stable over time, with monthly rates ranging from 4.00% to 5.16%. The largest month-to-month change was from March 2020 to April 2020 when the rate decreased from 5.09% to 4.00%. In June 2020, the rate had increased to 5.12%.

Figure 4. Monthly Combined Abdomen CT Rates



[7] Abdomen CT counts for December 2021 commercial data include only claims processed by December 31, 2021.